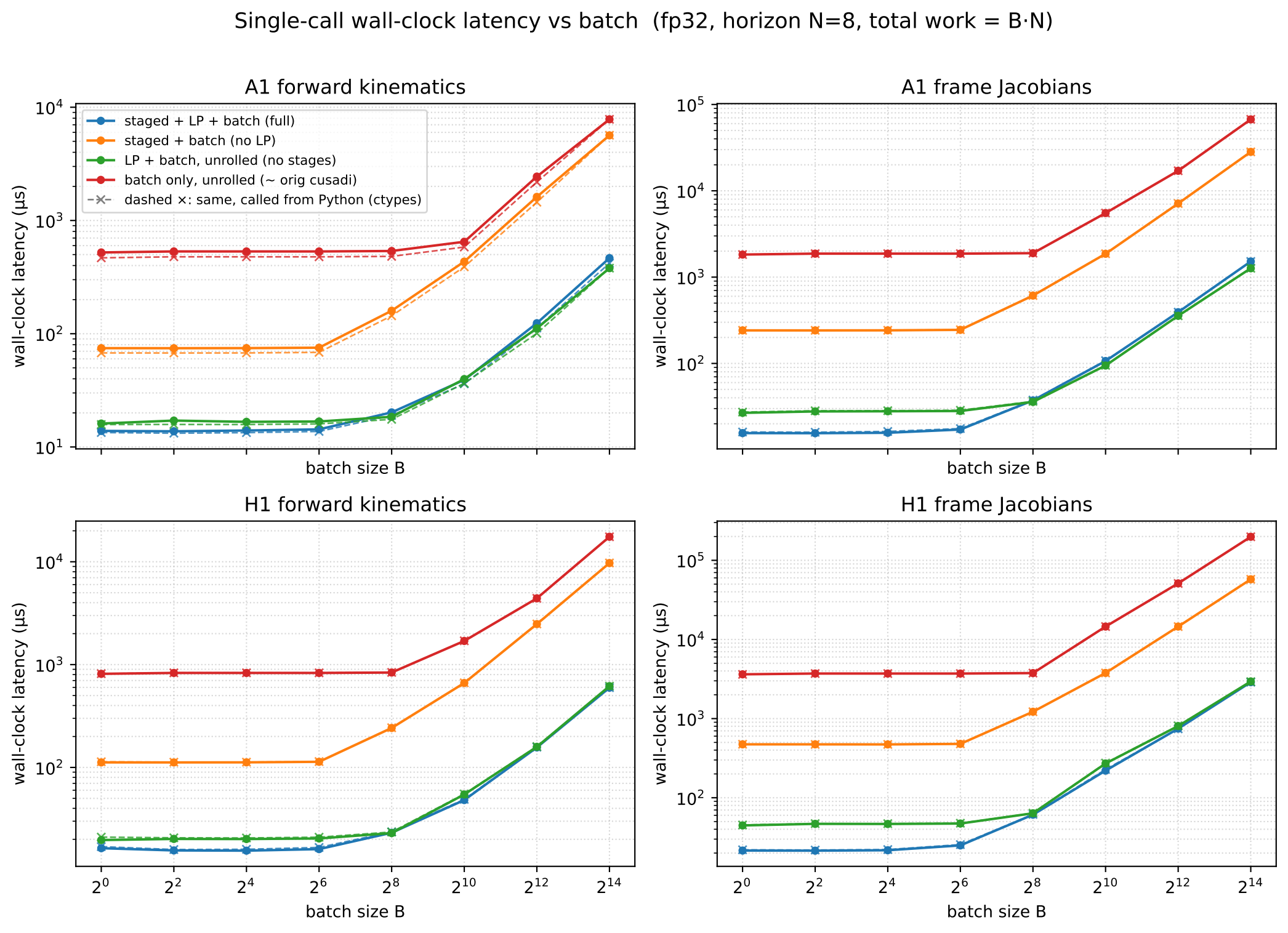
Single-call wall-clock latency vs batch (fp32, horizon N=8, total work = B·N)
A1 forward kinematics
A1 frame Jacobians
H1 forward kinematics
H1 frame Jacobians
staged + LP + batch (full)
staged + batch (no LP)
LP + batch, unrolled (no stages)
batch only, unrolled (~ orig cusadi)
dashed ×: same, called from Python (ctypes)
wall-clock latency (µs)
batch size B